Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 2: Euclidean and Extended Euclidean Algorithms

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Integer Arithmetic

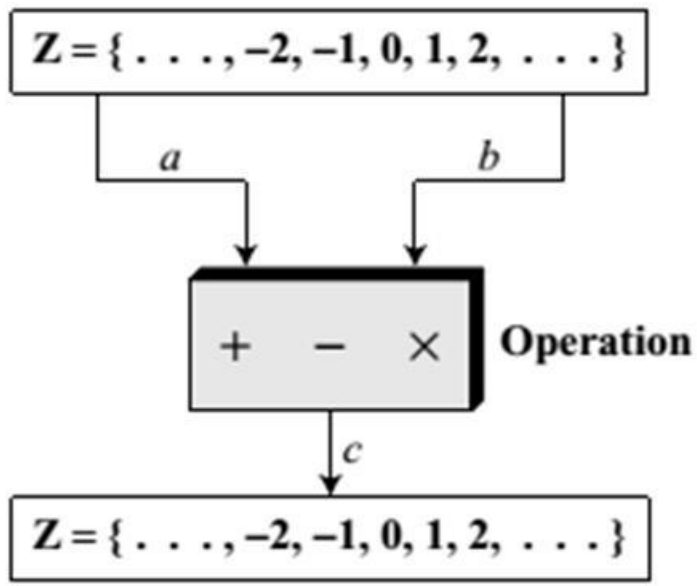
- In integer arithmetic, we use a set and a few operations.
- Reviewed here to create a background for modular arithmetic.
- The set of integers, denoted by Z, contains all integral numbers (with no fraction) from negative infinity to positive infinity

$$\mathbf{Z} = \{ \ldots, -2, -1, 0, 1, 2, \ldots \}$$

The set of integers

Binary Operations

- In cryptography, we are interested in three binary operations applied to the set of integers.
- A binary operation takes two inputs and creates one output.



Example #1

Add:	$5 + 9 = 14$	$(-5) + 9 = 4$	$5 + (-9) = -4$	$(-5) + (-9) = -14$
Subtract:	$5 - 9 = -4$	$(-5) - 9 = -14$	$5 - (-9) = 14$	$(-5) - (-9) = +4$
Multiply:	$5 \times 9 = 45$	$(-5) \times 9 = -45$	$5 \times (-9) = -45$	$(-5) \times (-9) = 45$

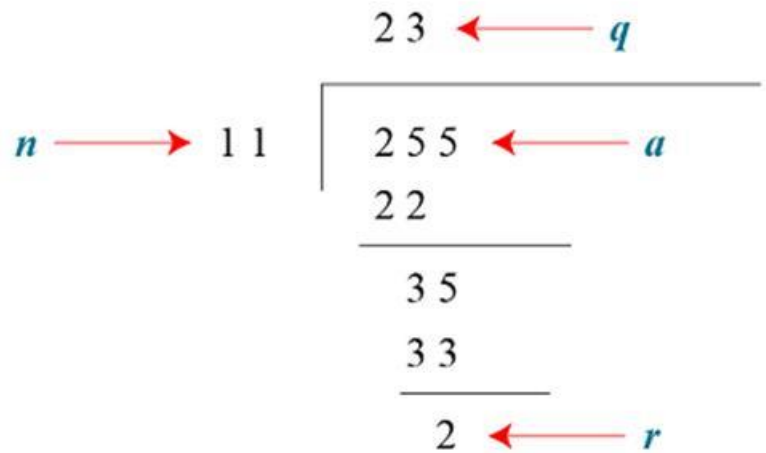
Integer Division

- In integer arithmetic, if we divide a by n, we can get q and r.
- The relationship between these four integers can be shown as

$$a = q \times n + r$$

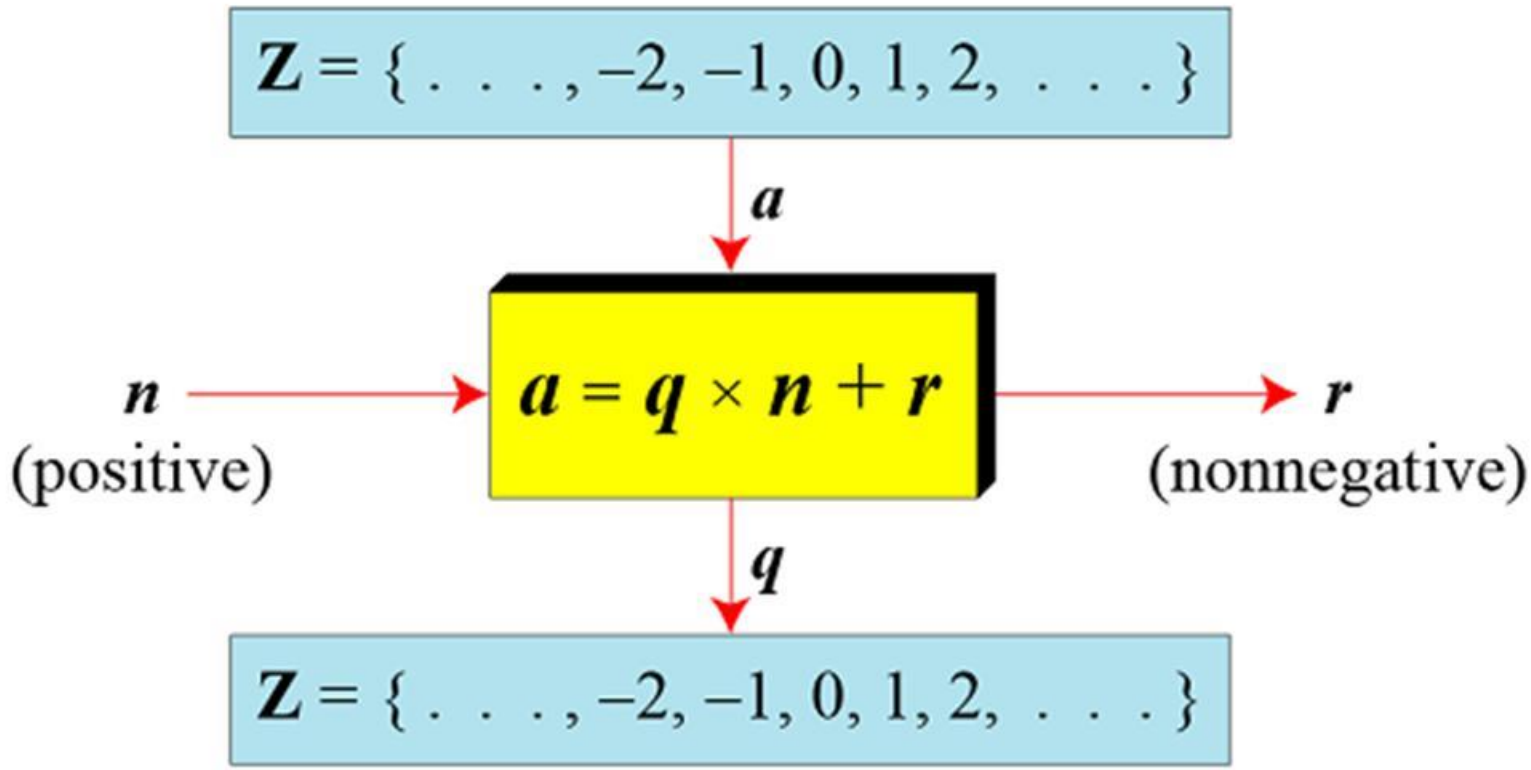
Assume that a = 255 and n = 11. We can find q = 23 and r = 2 using the division algorithm.

Example #2



Finding the quotient and the remainder

Integer Division



Division algorithm for integers

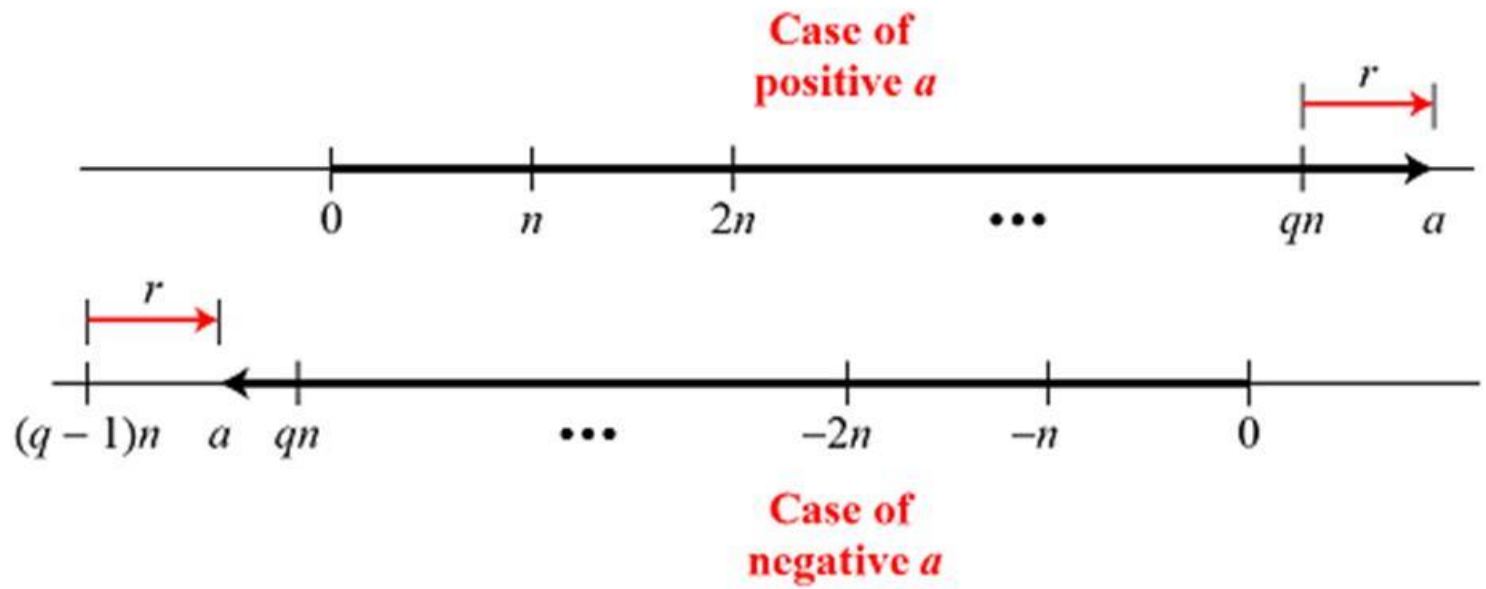
Integer Division

- When we use a computer or a calculator, r and q are negative when a is negative.
- How can we apply the restriction that r needs to be positive?
- The solution is simple, we decrement the value of q by 1 and we add the value of n to r to make it positive.

$$-255 = (-\mathbf{23} \times 11) + (-\mathbf{2}) \quad \leftrightarrow \quad -255 = (-\mathbf{24} \times 11) + \mathbf{9}$$

Integer Division

- Graph of division algorithm



Divisibility

- If a is not zero and we let r = 0 in the division relation, we get

$$a = q \times n$$

- If the remainder is zero, $n \mid a$
- If the remainder is not zero, $n \nmid a$

Example #3

a. The integer 4 divides the integer 32 because $\mathbf{32} = 8 \times \mathbf{4}$. We show this as 4|32.

b. The number 8 does not divide the number 42 because $\mathbf{42} = 5 \times \mathbf{8} + 2$. There is a remainder, the number 2, in the equation. We show this as $8 \nmid 42$.

Divisibility

- Properties

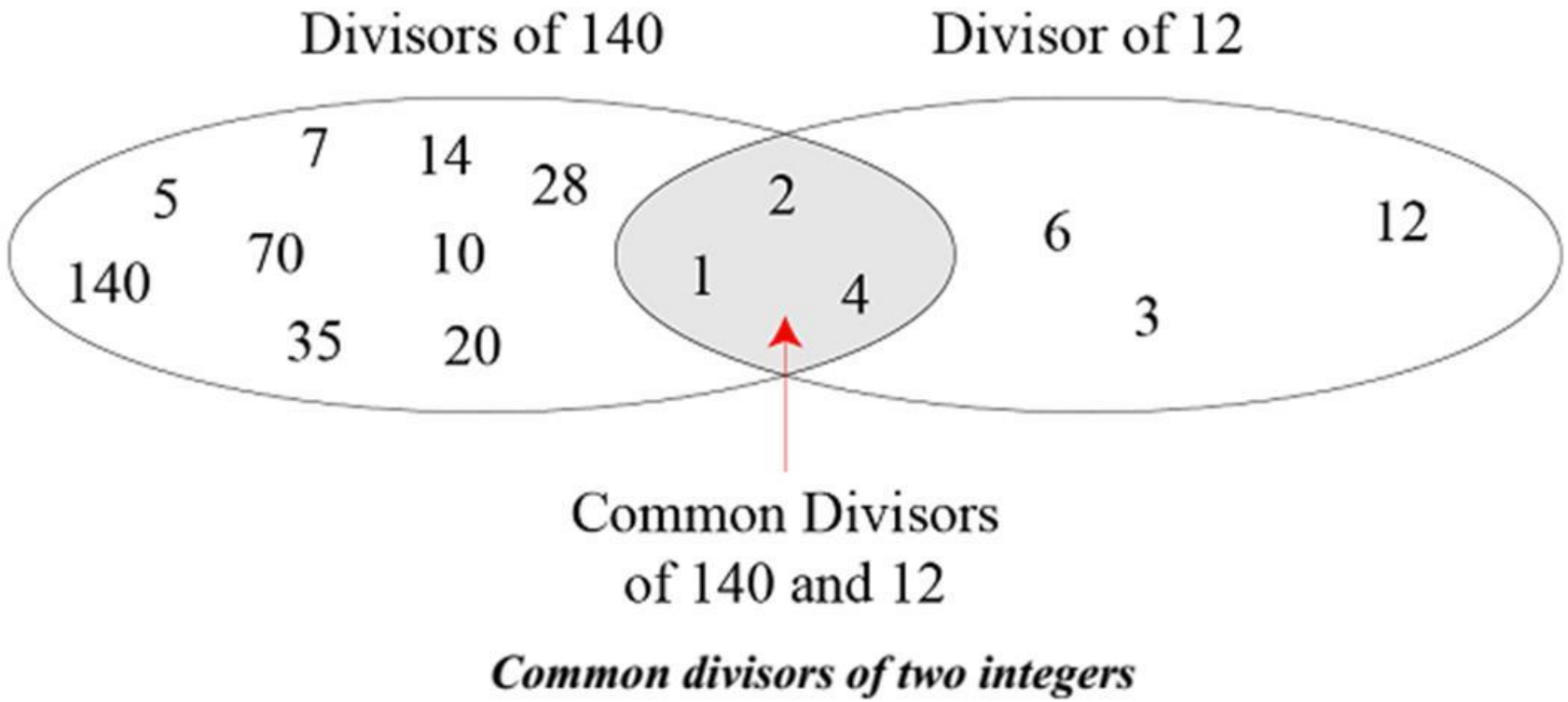
Property 1: if $a|1$, then $a = \pm 1$.

Property 2: if $a|b$ and $b|a$, then $a = \pm b$.

Property 3: if $a|b$ and $b|c$, then $a|c$.

Property 4: if $a|b$ and $a|c$, then $a|(m \times b + n \times c)$, where m and n are arbitrary integers

Divisibility



Common divisors of two integers

Divisibility

Greatest Common Divisor

The greatest common divisor of two positive integers is the largest integer that can divide both integers.

Euclidean Algorithm

Fact 1: gcd (a, 0) = a
Fact 2: gcd (a, b) = gcd (b, r), where r is the remainder of dividing a by b

Divisibility

- For example, to calculate the gcd(36,10), we use following steps:

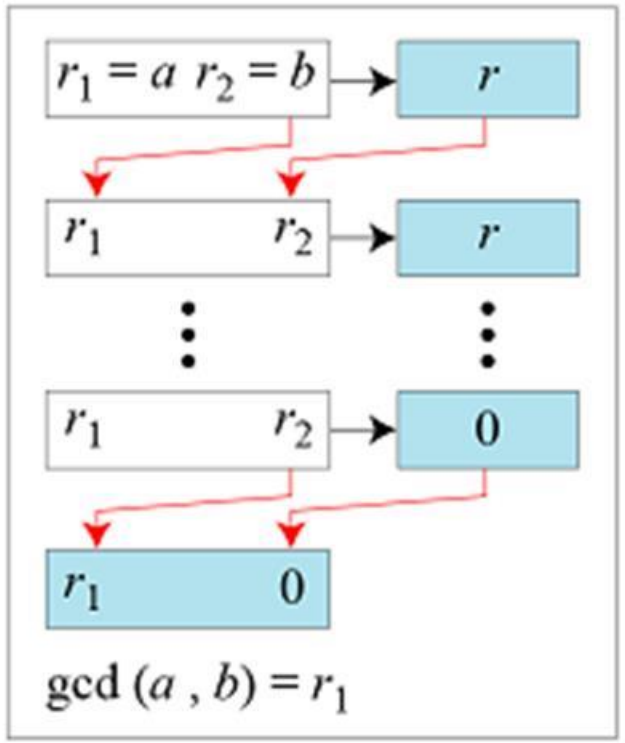
gcd (36, 10) = gcd (10, 6).....by fact 2
gcd (10, 6) = gcd (6, 4).....by fact 2
gcd (6, 4) = gcd (4, 2).....by fact 2
gcd (4, 2) = gcd (2, 0).....by fact 2
gcd (2, 0) = 2.....by fact 1

Hence, Answer = 2

Example #4

Divisibility

Euclidean Algorithm



a. Process

```
r1 ← a;      r2 ← b;        (Initialization)
while (r2 > 0)
 {
  q ← r1 / r2;
  r  ← r1 − q × r2;
  r1 ← r2 ; r2 ← r ;
 }
gcd (a, b) ← r1
```

b. Algorithm

When gcd (a, b) = 1, we say that a and b are relatively prime.

Divisibility

Find the greatest common divisor of 2740 and 1760.

q	r_1	r_2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

Example #5

Answer: gcd (2740, 1760) = 20.

Divisibility

Class Exercise #1

Find the greatest common divisor of 25 and 60.

Divisibility

- **Extended Euclidean Algorithm**

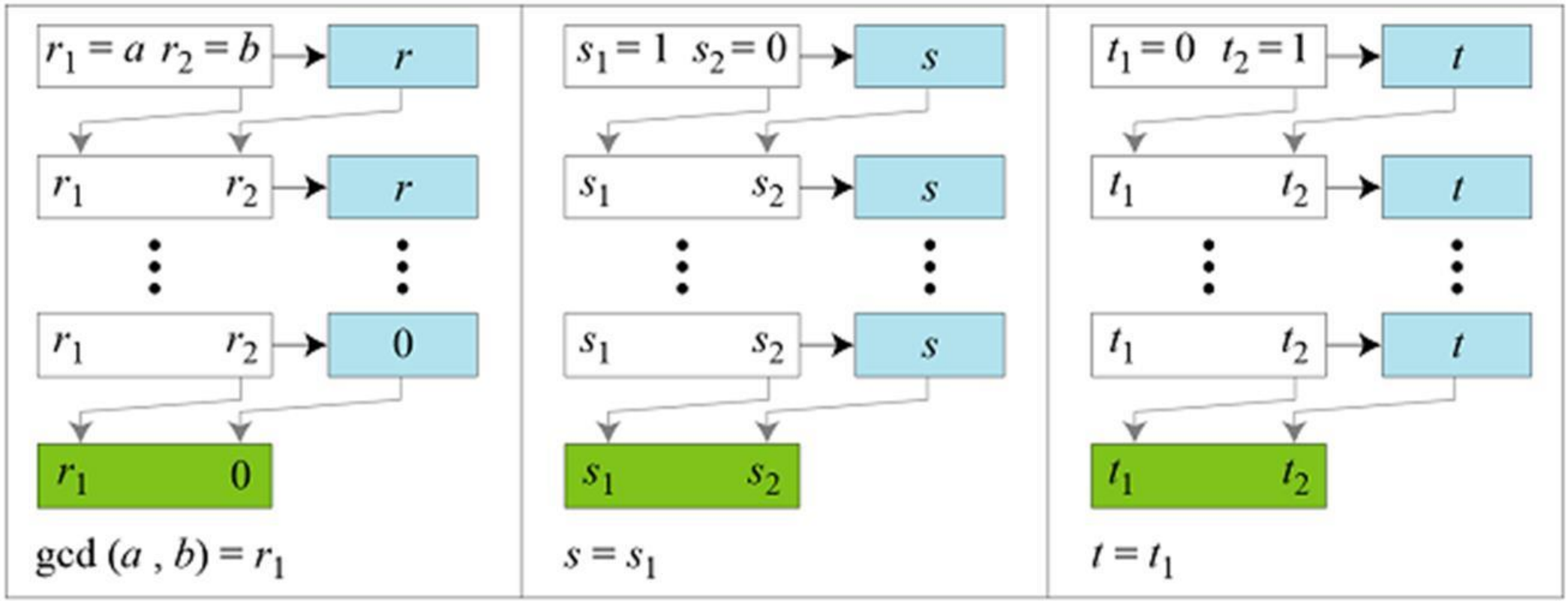
Given two integers *a* and *b*, we often need to find other two integers, *s* and *t*, such that

$$s \times a + t \times b = \gcd(a, b)$$

The extended Euclidean algorithm can calculate the gcd (a, b) and at the same time calculate the value of s and t.

Divisibility

Extended Euclidean algorithm, part a



a. Process

Divisibility

Extended Euclidean algorithm, part b

$r_1 \leftarrow a;$ $r_2 \leftarrow b;$
$s_1 \leftarrow 1;$ $s_2 \leftarrow 0;$
$t_1 \leftarrow 0;$ $t_2 \leftarrow 1;$ (Initialization)

while $(r_2 > 0)$

{

$q \leftarrow r_1 / r_2;$

$r \leftarrow r_1 - q \times r_2;$
$r_1 \leftarrow r_2;$ $r_2 \leftarrow r;$ (Updating r's)

$s \leftarrow s_1 - q \times s_2;$
$s_1 \leftarrow s_2;$ $s_2 \leftarrow s;$ (Updating s's)

$t \leftarrow t_1 - q \times t_2;$
$t_1 \leftarrow t_2;$ $t_2 \leftarrow$ t; (Updating t's)

}

gcd $(a, b) \leftarrow r_1;$ $s \leftarrow s_1;$ $t \leftarrow t_1$

b. Algorithm

Divisibility

Given $a = 161$ and $b = 28$, find gcd (a, b) and the values of s and t.

Solution

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	161	28	21	1	0	1	0	1	−5
1	28	21	7	0	1	−1	1	−5	6
3	21	7	0	1	−1	4	−5	6	−23
	7	0		**−1**	4		**6**	−23	

We get gcd (161, 28) = 7, $s = -1$ and $t = 6$.

Divisibility

Given $a = 17$ and $b = 0$, find gcd (a, b) and the values of s and t.

Solution

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
	17	0		**1**	**0**		**0**	**1**	

We get gcd (17, 0) = 17, $s = 1$, and $t = 0$

Divisibility

Given $a = 0$ and $b = 45$, find gcd (a, b) and the values of s and t.

Solution

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
0	0	45	0	1	0	1	0	1	0
	45	0		0	1		**1**	0	

We get gcd (0, 45) = 45, $s = 0$, and $t = 1$.

Divisibility

Exercise:

Given $a = 84$ and $b = 320$, find gcd (a, b) and the values of s and t.

Solution:

gcd(84,320) = 4, s = -19, t = 5

Assignment #1

- Explore and Investigate the following and submit a report by clearly mentioned your name and registration number in MS Word
  - **WhatsApp End to End Encryption Algorithm**
  - **Security and Privacy Features in WhatsApp**